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jakka_bhanu@LAPTOP-DCEV: ~$ cd ~/practical3
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ ls
process_states  process_states.c
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ gcc process_states.c -o process_states
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ ./process_states & sleep 2 && ps -elf | grep process_states && cat /proc/$(pgrep -f process_states | head -1)/status | grep State
[1] 532
=== Before fork() ===
Single process running. PID = 532, PPID = 347

[PARENT] Created child. Child PID = 533, My PID = 532, My PPID = 347
[PARENT] Sleeping for 3 seconds before checking on child...

[CHILD] Just created. PID = 533, PPID = 532
[CHILD] Sleeping for 10 seconds (state should show as 'S' = sleeping/waiting)...
jakka_b+ 532 347 532 0 1 05:15 pts/0 00:00:00 ./process_states
jakka_b+ 533 532 533 0 1 05:15 pts/0 00:00:00 ./process_states
jakka_b+ 536 347 536 0 1 05:15 pts/0 00:00:00 grep --color=auto process_states
State: S (sleeping)
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ [PARENT] Now calling wait() - I will block here (state 'S') until child terminates.
[CHILD] Woke up. Doing some work now (state should show as 'R' = running)...
[CHILD] Finishing now. PID = 533 will become a zombie briefly until parent calls wait().
[PARENT] Child has terminated and been reaped. Exiting now.
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$

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[PARENT] Finishing now. PID = 533 will become a zombie briefly until parent calls wait().
[PARENT] Child has terminated and been reaped. Exiting now.
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ ./process_states &
[2] 568
[1] Done ./process_states
=== Before fork() ===
Single process running. PID = 568, PPID = 347

[PARENT] Created child. Child PID = 570, My PID = 568, My PPID = 347
[PARENT] Sleeping for 3 seconds before checking on child...

[CHILD] Just created. PID = 570, PPID = 568
[CHILD] Sleeping for 10 seconds (state should show as 'S' = sleeping/waiting)...
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ [PARENT] Now calling wait() - I will block here (state 'S') until child terminates.
[CHILD] Woke up. Doing some work now (state should show as 'R' = running)...
[CHILD] Finishing now. PID = 570 will become a zombie briefly until parent calls wait().
[PARENT] Child has terminated and been reaped. Exiting now.

top - 05:18:38 up 32 min, 1 user, load average: 0.09, 0.07, 0.13
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7692.2 total, 6995.8 free, 619.9 used, 229.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7072.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24312   15400  11600 S   0.0   0.2   0:00.71 systemd
    2 root        20   0   3180    2212   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0   3180    2044   1940 S   0.0   0.0   0:00.00 init
   47 root       19  -1  42204   16600  15328 S   0.0   0.2   0:00.16 systemd-journal
   79 systemd+    20   0   22416   14572  12072 S   0.0   0.2   0:00.07 systemd-resolve
   90 root        20   0   35564   12504   9212 S   0.0   0.2   0:00.36 systemd-udevde
  167 root        20   0   2888    1952   1820 S   0.0   0.0   0:00.05 chronyd-starter
  168 root        20   0   4472    3100   2852 S   0.0   0.0   0:00.01 cron

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MiB Mem : 7692.2 total, 6995.8 free, 619.9 used, 229.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7072.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24312   15400  11600 S   0.0   0.2   0:00.71 systemd
    2 root        20   0   3180    2212   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0   3180    2044   1940 S   0.0   0.0   0:00.00 init
   47 root       19  -1  42204   16600  15328 S   0.0   0.2   0:00.16 systemd-journal
   79 systemd+    20   0   22416   14572  12072 S   0.0   0.2   0:00.07 systemd-resolve
   90 root        20   0   35564   12504   9212 S   0.0   0.2   0:00.36 systemd-udevde
  167 root        20   0   2888    1952   1820 S   0.0   0.0   0:00.05 chronyd-starter
  168 root        20   0   4472    3100   2852 S   0.0   0.0   0:00.01 cron
  169 message+    20   0   8796    5352   4672 S   0.0   0.1   0:00.05 dbus-daemon
  176 root        20   0   44096   29864  14724 S   0.0   0.4   0:00.26 networkd-dispat
  180 root        20   0   18560    9536   8232 S   0.0   0.1   0:00.09 systemd-logind
  183 root        20   0 1866032  144444  11576 S   0.0   0.2   0:00.13 wsl-pro-service
  199 syslog      20   0  220548    5596   4660 S   0.0   0.1   0:00.08 rsyslogd
  249 root        20   0   5492    2700   2516 S   0.0   0.0   0:00.00 agetty
  250 _chrony     20   0   23732   11184   7148 S   0.0   0.1   0:00.07 chronyd
  261 root        20   0 1234600  32692  18072 S   0.0   0.4   0:00.17 unattended-upgr
  265 _chrony     20   0   12096   24444   1804 S   0.0   0.0   0:00.01 chronyd
  345 root        20   0   3184   1116    980 S   0.0   0.0   0:00.00 SessionLeader
  346 root        20   0   3200   1252   1108 S   0.0   0.0   0:00.00 Relay(347)
  347 jakka_b+    20   0   6784    6044   3848 S   0.0   0.1   0:00.07 bash
  348 root        20   0   8648    5540   4696 S   0.0   0.1   0:00.00 login
  392 jakka_b+    20   0   22340   13420  10808 S   0.0   0.2   0:00.11 systemd
  396 jakka_b+    20   0   22912   4132   2132 S   0.0   0.1   0:00.00 (sd-pam)
[2]+ Done ./process_states
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ ./process_states & sleep 1 && echo "---- SNAPSHOT 1 (early) ----" && ps -elf | grep process_s
tates && sleep 5 && echo "---- SNAPSHOT 2 (mid-sleep) ----" && ps -elf | grep process_states && sleep 6 && echo "---- SNAPSHOT 3 (after
wake, near end) ----" && ps -elf | grep process_states
[1] 602
=== Before fork() ===

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jakka_bhanu@LAPTOP-DCEV: x + v
[1] 602
=== Before fork() ===
Single process running. PID = 602, PPID = 347

[PARENT] Created child. Child PID = 604, My PID = 602, My PPID = 347
[PARENT] Sleeping for 3 seconds before checking on child...

[CHILD] Just created. PID = 604, PPID = 602
[CHILD] Sleeping for 10 seconds (state should show as 'S' = sleeping/waiting)...
--- SNAPSHOT 1 (early) ---
jakka_b+ 602 347 602 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 604 602 604 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 606 347 606 0 1 05:18 pts/0 00:00:00 grep --color=auto process_states
[PARENT] Now calling wait() - I will block here (state 'S') until child terminates.
--- SNAPSHOT 2 (mid-sleep) ---
jakka_b+ 602 347 602 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 604 602 604 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 611 347 611 0 1 05:18 pts/0 00:00:00 grep --color=auto process_states
[CHILD] Woke up. Doing some work now (state should show as 'R' = running)...
[CHILD] Finishing now. PID = 604 will become a zombie briefly until parent calls wait().
[PARENT] Child has terminated and been reaped. Exiting now.
--- SNAPSHOT 3 (after wake, near end) ---
jakka_b+ 614 347 614 0 1 05:19 pts/0 00:00:00 grep --color=auto process_states
[1]+ Done ./process_states
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ ls /proc/$(pgrep -f process_states | head -1)/
1 2 348 90 cpuinfo fb kcore loadavg net softirqs tty
167 249 392 acpi crypto filesystems key-users locks pagetypeinfo stat uptime
168 250 396 buddyinfo devices fs keys meminfo partitions swaps version
169 261 422 bus diskstats interrupts kmsg misc pressure sys vmallocinfo
176 265 47 cgroups dma iomem kpagecgroup modules schedstat sysrq-trigger vmstat
180 345 627 cmdline driver ioports kpagecount mounts scsi sysvipc zoneinfo
183 346 7 config.gz dynamic_debug irq kpageflags mpt self thread-self
199 347 79 consoles execdomains kallsyms latency_stats mtrr slabinfo timer_list
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jakka_bhanu@LAPTOP-DCEV: x + v
[PARENT] Created child. Child PID = 604, My PID = 602, My PPID = 347
[PARENT] Sleeping for 3 seconds before checking on child...

[CHILD] Just created. PID = 604, PPID = 602
[CHILD] Sleeping for 10 seconds (state should show as 'S' = sleeping/waiting)...
--- SNAPSHOT 1 (early) ---
jakka_b+ 602 347 602 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 604 602 604 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 606 347 606 0 1 05:18 pts/0 00:00:00 grep --color=auto process_states
[PARENT] Now calling wait() - I will block here (state 'S') until child terminates.
--- SNAPSHOT 2 (mid-sleep) ---
jakka_b+ 602 347 602 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 604 602 604 0 1 05:18 pts/0 00:00:00 ./process_states
jakka_b+ 611 347 611 0 1 05:18 pts/0 00:00:00 grep --color=auto process_states
[CHILD] Woke up. Doing some work now (state should show as 'R' = running)...
[CHILD] Finishing now. PID = 604 will become a zombie briefly until parent calls wait().
[PARENT] Child has terminated and been reaped. Exiting now.
--- SNAPSHOT 3 (after wake, near end) ---
jakka_b+ 614 347 614 0 1 05:19 pts/0 00:00:00 grep --color=auto process_states
[1]+ Done ./process_states
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ ls /proc/$(pgrep -f process_states | head -1)/
1 2 348 90 cpuinfo fb kcore loadavg net softirqs tty
167 249 392 acpi crypto filesystems key-users locks pagetypeinfo stat uptime
168 250 396 buddyinfo devices fs keys meminfo partitions swaps version
169 261 422 bus diskstats interrupts kmsg misc pressure sys vmallocinfo
176 265 47 cgroups dma iomem kpagecgroup modules schedstat sysrq-trigger vmstat
180 345 627 cmdline driver ioports kpagecount mounts scsi sysvipc zoneinfo
183 346 7 config.gz dynamic_debug irq kpageflags mpt self thread-self
199 347 79 consoles execdomains kallsyms latency_stats mtrr slabinfo timer_list
jakka_bhanu@LAPTOP-DCEVSCQ0:~/practical3$ |
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